

Taylor Bosier

(402) 984-9277 · bosier83@gmail.com · tbosier.github.io

Summary

Lead Data Scientist with 6+ years of experience building and deploying production ML, Bayesian inference, and optimization models for high-impact business problems. Proven track record applying statistical modeling, supervised/unsupervised learning, and deterministic optimization to deliver measurable results in pricing, forecasting, and decision automation. Skilled in Python, SQL, and optimization solvers (Gurobi).

Professional Experience

Lead Data Scientist Stryker (Remote)

Aug 2024 – Present

- Built a causal inference pipeline using Databricks SQL and Python to evaluate rebate ROI with propensity score matching and inverse propensity weighting across customer-level rebate engagement and eligible-product revenue.
- Developed a long-horizon forecasting pipeline (several XGBoost models) used for strategic planning of 18-36 month revenue forecasting, with cold start handling.
- Built a Databricks Vector Search workflow that parallelized ingestion of 35,000+ contracts from Azure Data Lake Storage, converted PDFs into chunked text, used AI-rewritten semantic queries to retrieve rebate structures, tied complex rebate terms to sales data, and rendered rebate attainment progress in a Next.js graph.
- Developed a pricing benchmark engine using XGBoost trained on historical sales, producing median and 75th-percentile price estimation by item/customer/region/contract type to standardize “market rate” guidance and reflect top-performer pricing behavior.
- Led customer segmentation and product-mix analytics to support operational planning and identify high-value cohorts for targeted distribution strategies.
- Mentored senior data scientist and full-stack web developers and collaborated cross-functionally with finance, operations, and pricing teams to operationalize models.

Data Scientist II Knight-Swift Transportation, Phoenix, AZ

Sept 2021 – Aug 2024

- Promoted to Data Scientist II in Sept 2023.
- Re-architected and deployed a scheduling optimization model (Gurobi), improving on-time delivery rates and customer service reliability, and improving the network design.
- Developed ML models to optimize pricing strategies
- Explored reinforcement learning for schedule planning
- Designed an early-warning detection model to enhance driver safety and reduce risk
- Conducted a study using a Bayesian Poisson-Gamma model to quantify the safety impact of lane-keeping sensors, providing critical evidence for a multi-million dollar fleet hardware decision.

- Optimized a driver retention model, yielding a 20% increase in F1-score and leading to 12% drop in driver turnover.
- Performed frequency-severity modeling for risk assessment in insurance purchasing, leading to the purchase of \$60 million insurance coverage

Business Analyst

June 2019 – Sept 2021

Chain Link Services

- Determined optimal warehouse locations via k-means clustering and statistical analysis, reducing inventory costs and stockouts.
- Employed a Facebook Prophet model to forecast inventory demand, lowering holding costs and space requirements.
- Improved warehouse pick speeds by restructuring layout with an apriori model and implementing aisle-traveling heuristics.
- Deployed a Safety Stock algorithm ($Z\text{-score} \times \text{std. of lead time} \times \text{avg. demand}$) to set offsite warehouse inventory levels, cutting physical space and purchase costs.

Technical Skills and Competencies

- **Python:** PyMC, scikit-learn, PyTorch, XGBoost, PySpark
- **Optimization Modeling & Solvers:** Pyomo, PuLP, Gurobi, HiGHS
- **Simulation:** ARENA, SimPy
- **SQL:** Databricks, Microsoft SQL Server, DB2, Hadoop, Elasticsearch
- **Cloud:** Databricks, Azure, Cloudera Data Science Workbench
- **Version Control:** Github, GitLab
- **Orchestration:** Prefect, Airflow

Education

New Mexico State University

Las Cruces, NM

Master of Science in Mathematics (May 2018)

New Mexico State University

Las Cruces, NM

Bachelor of Science in Mathematics (May 2016)

Hobbies and Interests

- Math, coding, espresso
- Reading sci-fi/fantasy, caring for my 3 cats, exploring the city with my wife.